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KULLIYAH OF ENGINEERING

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

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**WEEK 7: PLC INTERFACING WITH MICROCONTROLLER AND PC OVER
ETHERNET/IP**

SECTION 18

GROUP 11

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ABSTRACT

This experiment focused on the development of a Start-Stop Control Circuit using OpenPLC and an Arduino board. The main objective of this experiment was to create, simulate, and implement a ladder diagram control system that allows an LED to be controlled using START and STOP push buttons. OpenPLC Editor was used to design and simulate the ladder logic program before transferring it to the Arduino board.

The system applied a latching concept, where the LED remains turned on after the START push button is pressed once. Without the latch function, the user would need to continuously press the START push button to keep the LED turned on. The STOP push button was used to break the electrical path and turn off the LED. The experiment successfully demonstrated the operation of a PLC-based control system and showed how ladder logic can be implemented for industrial automation applications. Overall, the objectives of the experiment were achieved successfully.

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INTRODUCTION

Programmable Logic Controllers (PLCs) are industrial computers widely used in automation and control systems. PLCs are designed to control machines and industrial processes efficiently, reliably, and safely. One of the most common programming methods used in PLC systems is ladder logic programming, which represents electrical control circuits in a graphical form. Ladder diagrams are widely used because they are simple to understand and suitable for industrial automation applications.

In this experiment, OpenPLC Editor was used to create and simulate a ladder diagram for a Start-Stop Control Circuit. The ladder logic program was then uploaded to an Arduino board, which acted as the PLC controller. The circuit consisted of two push buttons, namely the START and STOP buttons, and an LED as the output device. The START push button was used to activate the LED, while the STOP push button was used to deactivate it.

The experiment applied a latching concept in the control circuit. When the START push button was pressed once, the LED remained turned on even after the button was released. This was achieved by using a latch in the ladder logic program. Without the latch function, the user would need to continuously press the START push button to keep the LED on. The STOP push button was used to break the electrical path and turn off the LED. This experiment demonstrated the basic principles of PLC control systems and the importance of latching circuits in automation systems.

TASK : Development of Start-Stop Control Circuit using OpenPLC and Arduino

1.0 MATERIALS AND EQUIPMENT

- OpenPLC Editor software
- Cytron CT-UNO Board
- 2 Push Button Switches
- LED
- Resistors
- Jumper Wires
- Breadboard

2.0 EXPERIMENTAL SETUP

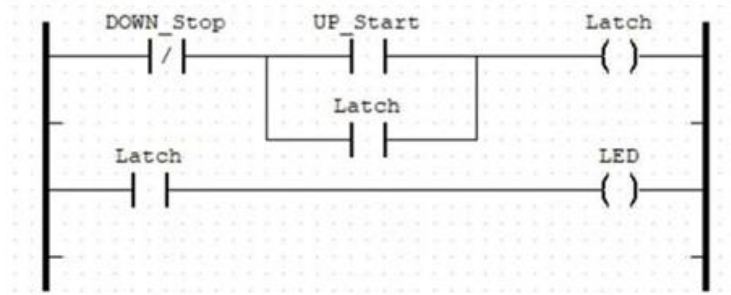


Fig 1.0 shows the ladder diagram in the OpenPLC Editor.

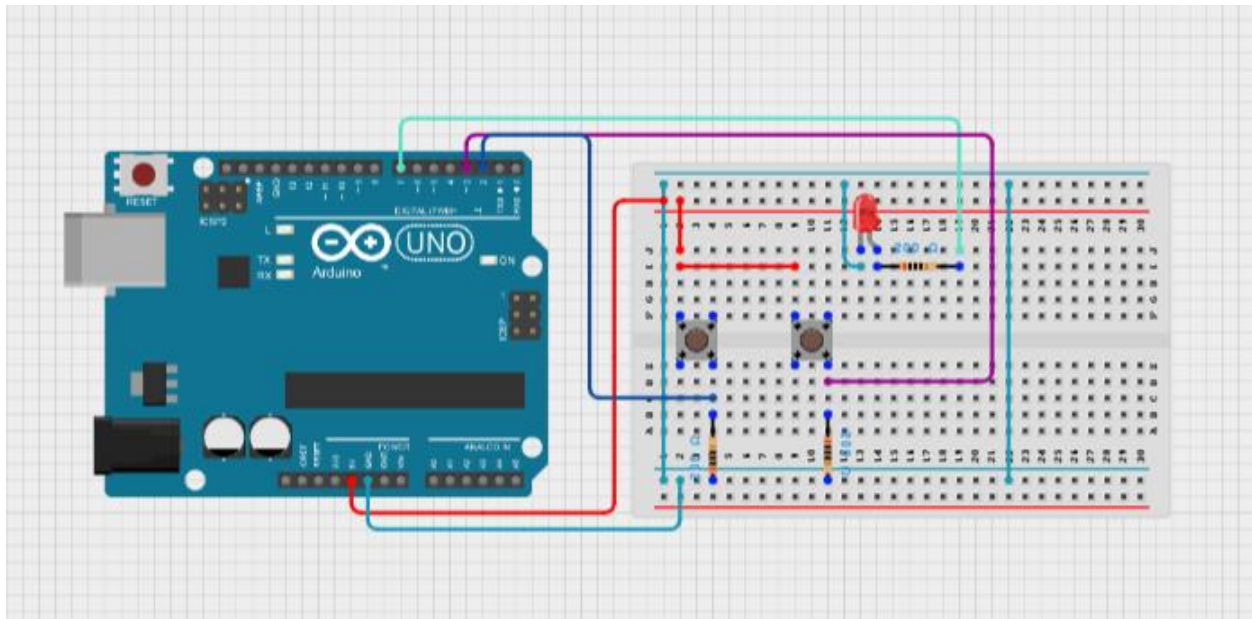


Fig 1.1 shows the connection for the task.

1. Create the ladder diagram as shown in Fig. 1.0 above.
2. Specify all variables used in the ladder diagram.
3. Compile and simulate the ladder diagram in OpenPLC Editor.
4. Upload the ladder diagram to the Arduino board.
5. Ensure to select the correct COM port number and all pin associations between the OpenPLC variables and Arduino board.
6. Build the circuit as illustrated in Fig. 1.1.
7. Test the functionality.

3.0 METHODOLOGY

a. Circuit Assembly

- The push button 1 (start button) is connected to the pin 2 of Cytron CT-UNO and the other leg of the start button is connected to the 5V.
- The stop button is connected to the pin 3 of Cytron CT-UNO and the other leg will connect to the 5V.
- The resistor 1 is connected to the pin 2 of Cytron CT-UNO and the other leg is connected to the GND.
- Resistor 2 is connected to the pin 3 of Cytron CT-UNO and the other leg is connected to the GND.
- The resistor 3 is connected to the pin 7 of Cytron CT-UNO and the other leg is connected to the GND.
- The cathode of the LED is connected to the GND and the anode is connected to the pin 7.

b. Programming and Uploading

- The Open PLC editor is being downloaded, and the software is being launched.
- Select File, New, New folder and name it as Select Folder to start a new project.
- Name the project and the language of ladder diagram is being selected.
- The symbol '+' is being pressed to create a new variable and name the file.
- Set the class filter as local and Type as BOOL on the table.
- Once the ladder diagram is finished, push the compile button.
- Set the location for each elements with correct location based on the connection with the Cytron CT-UNO.

c. Table

Description :

Class Filter : All

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#	Name	Class	Type	Location	Initial Value	Documentation	Debug
0	DOWN_STOP	Local	BOOL	%IX0.1			
1	UP_START	Local	BOOL	%IX0.0			
2	LATCH	Local	BOOL				
3	LED	Local	BOOL	%QX0.0			

Figure 1.2 shows the table for task 1 in PLC editor.

d. Ladder diagram

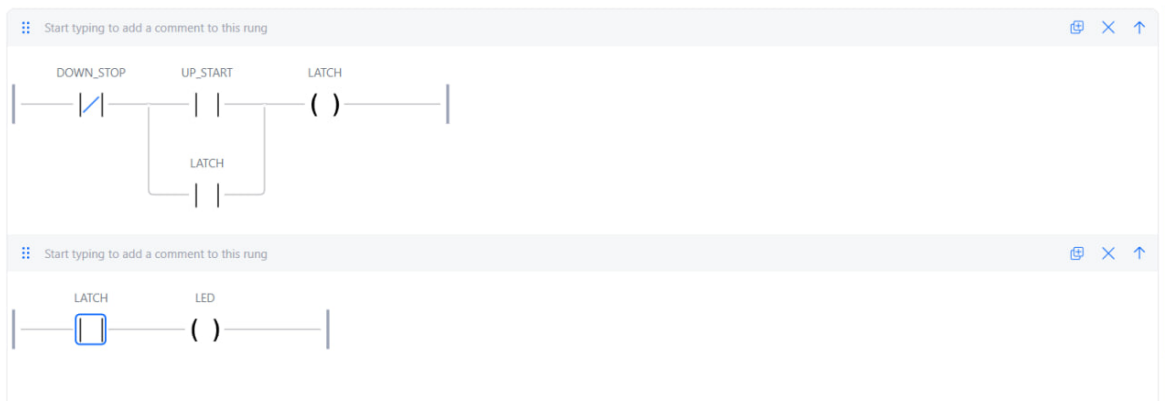


Figure 1.3 shows the table for task 1 in PLC editor

4.0 DATA ANALYSIS

START PUSH BUTTON	STOP PUSH BUTTON	LED
Being pressed	Not being pressed	ON
Not being pressed	Being pressed	OFF

When the start push button is being pressed and the stop push button is not being pressed, the led will turn on and when the stop push button is being pressed the LED will turn off. This data used a latching concept. Without having a latch on the circuit, the user needs to keep pressing the start push button to make the LED turn on. However, the latch on the circuit will help the user to only press the start push button once to turn on the LED. We can turn off the LED by pressing the stop button once because it will break the electrical path.

5.0 RESULTS

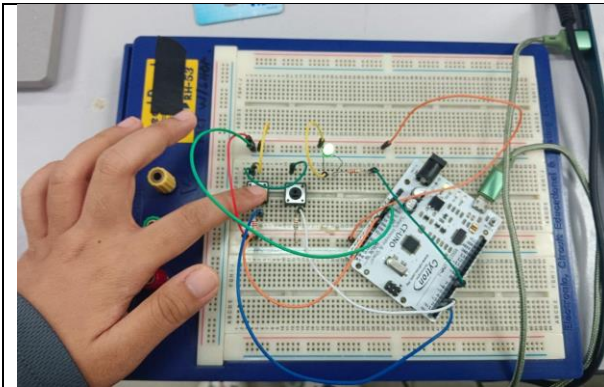


Figure 1.3 shows the start button is being pressed and the LED turns on.

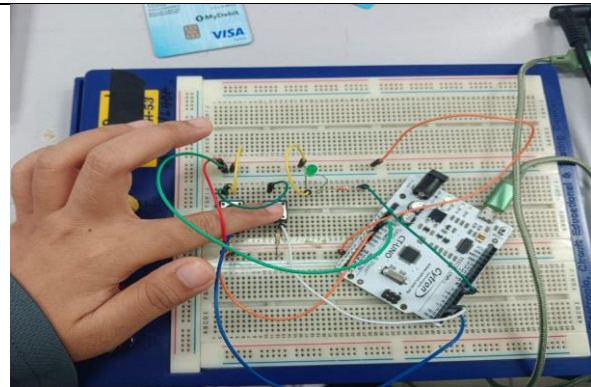


Figure 1.4 shows the stop button is being pressed and the LED turns off.

6.0 DISCUSSION

The Start-Stop Control Circuit was successfully developed and tested using OpenPLC and an Arduino board. Based on the experiment, the system functioned according to the designed ladder logic. When the START push button was pressed while the STOP push button was not activated, the LED turned on successfully. The LED remained on even after the START push button was released because the circuit applied a latching concept.

The latching concept played an important role in this experiment because it allowed the output to maintain its state without continuously pressing the START push button. Without the latch function, the user would need to hold the START button continuously to keep the LED turned on. The latch created a continuous electrical path in the ladder logic, which maintained the output condition until the STOP push button was pressed.

When the STOP push button was pressed, the electrical path was broken, causing the LED to turn off immediately. This demonstrated the operation of a basic industrial control system commonly used in automation applications. The experiment also showed that ladder logic programming is effective for designing simple control systems because it is easy to understand and closely resembles traditional electrical relay circuits.

During the experiment, proper pin configuration and correct COM port selection were important to ensure successful communication between OpenPLC and the Arduino board. Any

incorrect wiring or wrong pin association could cause the circuit to malfunction. Overall, the experiment provided a better understanding of PLC programming, ladder logic design, and the implementation of latching circuits in industrial automation systems.

7.0 CONCLUSION

In conclusion, the objectives of this experiment were successfully achieved. A Start-Stop Control Circuit was successfully designed, simulated, and implemented using OpenPLC and an Arduino board. The ladder logic program operated correctly, where the START push button turned on the LED and the STOP push button turned it off.

The experiment also demonstrated the importance of the latching concept in maintaining the output state without requiring the START push button to be continuously pressed. The STOP push button successfully broke the electrical path and deactivated the LED. Through this experiment, a better understanding of PLC systems, ladder logic programming, and industrial automation concepts was obtained. Overall, the experiment was successful and helped improve practical knowledge in PLC interfacing and control system development.

8.0 RECOMMENDATION

Several improvements can be made to enhance the performance and functionality of the Start-Stop Control Circuit. First, additional output devices such as buzzers, motors, or relays can be added to make the system more suitable for real industrial applications. Sensors may also be integrated into the system to allow automatic operation based on environmental conditions. According to Abduraimov (2023), a basic Start–Stop circuit can be industrialized by adding robust motor starters and protection (star–delta or current-limited drives, under voltage and fault relays), richer output devices (contactors, lamps, alarms), and a microcontroller/PLC layer. By integrating the appropriate sensors and closed-loop control, it will enable an automatic start–stop and speed adjustment based on environmental and process conditions, improving safety, reliability, and efficiency.

Besides that, a more advanced PLC hardware system can be used instead of a basic Arduino board to improve reliability and efficiency. According to Sehr et al. (2021), Industrial PLCs are modular, ruggedized, and provide a stable runtime with cycle-time monitoring and protections that prevent programming errors from compromising the control system. Proper cable management and stronger circuit connections should also be implemented to reduce connection errors during operation. Furthermore, safety features such as emergency stop buttons and protective components can be added to improve the safety of the control system.

9.0 REFERENCES

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10.0 ACKNOWLEDGEMENT

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11.0 STUDENTS DECLARATION

We hereby declare that, except for the places where it is acknowledged, all the work presented in this report is entirely ours. We certify that, in completing this project, we have

complied with the standards of academic integrity and have not engaged in any plagiarism or unethical behaviour. Every information and support source used in this work has been appropriately referenced and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein has not been done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributor to the report. We therefore agreed unanimously that this report shall be submitted for marking, and this final report has been verified by us.

Signature: <i>nurulanis</i> Name: NURUL ANIS BINTI MOHD KHAIRI Matric Number: 2419788	Read [/] Understand [/] Agree [/]
Signature: <i>athirahmeor</i> Name: WAN NUR ATHIRAH BINTI MEOR MOHD REDZUAN Matric Number: 2320102	Read [/] Understand [/] Agree [/]
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